

Marks –8

- The business requirements are fine and aligned to the learning outcomes.
- Good writing, has a flow, use good citations
- Good discussion about the first BR provided examples and discussed it. I suggest you that you could have improved it more by giving some detail information about data, providing some figures to strengthen your arguments such as a figure of a dashboard
- There is a good attempt at analysis, but the critical evaluation could be deepened by including more real-world examples or case studies
- Need a bit more about data that would make your writing more strong
- Try linking analysis findings to BR a bit detail

Criteria	10-9	8-7	6-5	4-0
The role of BI in the organisation	The report Judiciously demonstrates a critical evaluation of the role of advanced Business intelligence tools and techniques and includes: • an appropriate critical review of different business requirements (at least 3) within a specified NZ-based organisation • critical evaluation demonstrating the role of data analysis to address business requirements in a specific industry. and is supported by appropriate and judicious use of peer reviewed conference proceedings or journal articles.	The report clearly demonstrates a critical evaluation of the role of advanced Business intelligence tools and techniques and includes: • an appropriate critical review of different business requirements (at least 3) within a specified NZbased organisation • critical evaluation demonstrating the role of data analysis to address the business requirement in a specific industry. and is supported by the appropriate use of peer-reviewed conference proceedings or journal articles.	The report demonstrates a critical evaluation of the role of advanced Business intelligence tools and techniques and includes: • an appropriate critical review of different business requirements (at least 3) within a specified NZbased organisation • critical evaluation demonstrating the role of data analysis to address the business requirement in a specific industry. and is supported by the appropriate use of peer-reviewed conference proceedings or journal articles.	The report does not demonstrate a critical evaluation of the role of advanced Business intelligence tools and techniques and includes: • an appropriate critical review of different business requirements (at least 3) within specified NZbased organisation • critical evaluation demonstrating the role of data analysis to address the business requirement in a specific industry. and is not supported by the appropriate use of peer-reviewed conference proceedings or journal articles.

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Introduction

In recent times, the use of Big Data has enabled organizations to leverage business intelligence to analyze data to better understand its business and market to make factual business decisions. Using it in day-to-day activities with the help of big data has helped organizations find opportunities, risks, and support decision making (Niu et al., 2021).

Xero, a global leader in cloud-based accounting software, is a prime example of how BI can drive innovation and organizational success. Founded in 2006 in New Zealand, Xero provides small and medium-sized businesses with tools to manage their finances, including invoicing, payroll, and tax filing (Xero). For the organization Xero, BI could be utilized in various ways such as analyzing customer data to understand needs of accountants and small businesses, analyzing employee data to understand and monitor productivity and how it can improve employee satisfaction and reduce attrition. These helps in potentially identifying opportunities for new features or service improvements (Božič & Dimovski, 2019).

This report will discuss in detail three business requirements leveraging Business Intelligence at Xero. Which is then followed by conclusion which briefly discusses the business requirements, limitations and recommendations.

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Business Requirements for Xero

Interactive Visualization for Monitoring Employee Performance

In Xero, business intelligence can be leveraged to improve or monitor employee performance. But what is employee performance? It is the level of success employees achieve in carrying out their duties and responsibilities. Employee performance is key to achieving organizational goals and significantly impacts company output. Effectively monitoring employee performance is a core business requirements for most organizations and achieving organizational goals is critically dependent on this performance. (Tamunomiebi & Oyibo, 2020). Due to this, it has a chain effect on how we deliver to customers. Monitoring can provide employees with immediate feedback on their achievements and areas needing improvement, helping them understand how much work is needed and where to focus resources (Sherif et al., 2021). It helps managers assess if objectives are being met and can support decisions regarding intervention on individual performance or actions for underperformers(Pape, 2016).

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The process of monitoring involves defining expectations, define rules, collecting, assessing, and providing feedback on employee behavior and job performance to drive improvement (Audenaert et al., 2019). Common tools used for monitoring employee performance are Employee Performance Management (EPM), AI Applications, and Business Intelligence (BI) systems. EPM, which involves using technology (video monitoring, call monitoring, time clock system, etc.) to observe, record, analyze information related to job performance (Ravid et al., 2020). It is defined as use of tools aimed at focusing employees on organizational goals (Sherif et al., 2021). AI Applications is the use of big data analytics and self-learning capabilities to provide job evaluation and performance feedback to employees (Tong et al., 2021). BI systems is the use of various techniques to integrate information, like visualization, to gather and analyze data on various topics like current performance vs expectation (Richards et al., 2019).

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For the techniques under BI, there are various techniques such as regression, clustering, visualization, etc.... Visualization is used to present data and insights into visual formats such as dashboards, charts and graphs (Gusnadi & Hermawan, 2019). Dashboards can be a great tool in showing the comparison of employee performance within a team/department in Xero. While regression can be used to predict employee performance, determine factors that affect productivity, helping organizations understand factors that

impact performance (Tak, 2024), it is challenging to translate findings into widespread understanding and action across a company (Božič & Dimovski, 2019). And while clustering can be used to identify distinct groups in Xero to provide tailored strategies like incentive systems or training programs for each group (PM & Balaji, 2019), similar to regression, the raw output of clustering needs interpretation to be meaningful. Visualizations will be used in this report, due to its ability to turn raw numbers into easy to understand format (Kobi, 2024).

Under BI systems, while there are many tools that can be used to monitor employee performance like Tableau, PowerBI, Jira, etc. with each tool having their own strengths and weaknesses against other tools. This report will focus more on PowerBI because even though other tools such as Jira which is more focused on task management and has reports(sprint velocity, sprint breakdown chart, etc.) (Sokolova, 2024), and Tableau featuring more complex features (Karl, 2024). PowerBI will be used because it is flexible, simpler, and much more suited for smaller to medium sized data (Biswal, 2025), which is suited to this report.

For the data that will be used in visualization, employee data such as number of tasks finished, and work quality, feedback etc. (Gusnadi & Hermawan, 2019; Zhao et al., 2019), which will be used to create the dashboard for displaying employee performance at Xero. The data collected will be analyzed, which the outputs include calculated results for key performance indicators (KPIs), performance against expectation, employee ranking (Mellisa, 2019; Zhao et al., 2019). These results has many benefits for Xero: potential to incentivize effort and performance, improve workforce management, productivity, employee engagement, and optimize work balance (Ravid et al., 2020; Tak, 2024; Tong et al., 2021; Zhao et al., 2019).

Predicting Employee Attrition using Logistic Regression

Not just in Xero, but employee attrition has been an issue in most organizations. Employee attrition, also known as employee turnover, is defined as the process where employees leave an organization for various reasons. Aside from loss of employees, employee attrition affects other aspects such as loss of valuable, high performing employees, and cost in professional training, recruitment, and hiring (Qutub et al., 2021). According to Raza et al., (2022), the average cost per hire for a new employee is around 4129 US dollars.

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Using business intelligence tools can be leveraged to address or reduce employee attrition effectively in Xero by gathering data-driven insights.

To effectively address employee attrition, there are many techniques that we can use such as clustering, prediction, decision trees, etc. (Alsheref et al., 2022; Jain et al., 2020; Yang & Islam, 2020). Each of them has their own strengths and weaknesses. Using clustering, Xero can classify which type of employees who are more likely to quit based on risk levels (high risk, low risk) (Alsheref et al., 2022). Organizations can understand the common profiles of employees who are prone to leaving (e.g. younger, further commute, lower income, etc.). However, clustering does not fully capture the various reasons behind an employee's intention to leave (Yang & Islam, 2020). Using prediction on the other hand, Xero can determine whether the probability that the employee is likely to stay or leave, this is intended to help intervene proactively, develop retention strategies, and potentially prevent valuable employees from leaving (Qutub et al., 2021). However, predicting employee attrition is heavily reliant on data, and these data are often internal to companies, confidential, and difficult to obtain (Jain et al., 2020). Finally, using decision trees, which is similar to prediction, Xero can classify individual employees who are likely to leave or stay based on their features (El-Rayes et al., 2020). However, decision trees does not always achieve the highest accuracy compare to other complex methods (Raza et al., 2022).

This report will utilize prediction since our aim is to proactively identify employees who are likely to leave Xero and understand factors influencing this decision. There are many tools that can be used to predict employee attrition such as Logistic Regression, Naïve Bayes, Random Forest, etc.... Logistic regression is a great algorithm used for predicting the relationship between employee characteristics and attrition (binary outcome). It showed better numbers in terms of accuracy, true positives, etc.... (Qutub et al., 2021). However, accuracy in other studies proved to vary, from 72% to 81% in multiple studies (Fallucchi et al., 2020; Mhatre et al., 2020; Raza et al., 2022; Setiawan et al., 2020). Naïve Bayes, praised for its superior ability to identify true positives, which is essential for identifying the largest number of employees at risk of leaving. However, its overall accuracy or precision might be lower in some comparisons (Fallucchi et al., 2020). Random forest, a method based on decision trees, is often cited as one best performing algorithms for employee attrition, achieving high overall accuracy in multiple studies (Bhartiya et al., 2019; Jain et al., 2020; Raza et al., 2022; Yang & Islam, 2020). Random Forest has a strong explanatory nature and can calculate the importance of each variable, which is important in identifying factors that

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leads to attrition (Yang & Islam, 2020). However while often high in accuracy, it can be outperformed by other tools like Logistic Regression in some specific datasets (Setiawan et al., 2020). It is much more complex than other models, offering only marginal gains (El-Rayes et al., 2020).

For this report, Logistic Regression will be used as it is more suitable for probability prediction that is focused on binary classification(resign, stay) (El-Rayes et al., 2020). For the data that will be used in this report, the employee's details such as age, gender, education, employee satisfaction, income, seniority, and demographics will be used (Mansor et al., 2021). Work details will also be used such as work environment, work-life balance, job involvement, job level, job role, job satisfaction, relationship satisfaction, training times, total working years, years at company, years in current role, years since last promotion, years with current manager (Pratt et al., 2021). The result of this prediction will provide early detection on the employees who are likely to resign, this allows the company to intervene proactively, resulting in retention of valuable employees (Qutub et al., 2021).

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Improving Employee Engagement and Satisfaction using K-Means

For a company like Xero, employee engagement and satisfaction are essential factors as it affects other aspects of the company. It has shown that it has positive relationship with customer satisfaction, employee retention, employee performance (Kang et al., 2020).

To effectively improve employee engagement and satisfaction, there are many business intelligence techniques such as prediction, clustering, statistical analysis, etc.... Each has their own strengths and weaknesses. Prediction can predict satisfaction levels with high accuracy and provide early detection for companies. However, there are challenges such as finding the best performing prediction tool (Conlon et al., 2021). Clustering can reveal patterns or factors specific to each group that are associated with job satisfaction. One study showed that one cluster might consist of younger employees with many awards and high salaries who are certified and satisfied (Christianto et al., 2023). However, clustering can be challenging, it can result to too many clusters. This can influence interpretability (Christianto et al., 2023). Statistical analysis, on the other hand, can analyze how certain attributes (experience, gender, age, etc.) relate to satisfaction and engagement levels. This helps identify if certain groups have different satisfaction drivers (Rashidin et al., 2020). However,

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while statistical analysis can show relationships between variables, it does not necessarily prove that one factor causes another (Zhang et al., 2020).

This report will utilize clustering since our goal is to understand the different types of employees and tailor strategies to their unique needs and drivers of satisfaction (Christianto et al., 2023). There are many clustering tools that we can use to improve employee engagement and satisfaction such as K-Means, Hierarchical Clustering, K-Medoid Clustering, etc.... K-Means is a simple method that is useful in grouping employees into groups based on characteristics such as satisfaction levels (very satisfied, moderately satisfied, not satisfied, etc.) (Sekhon & Sadawarti, 2021). However, the result can be difficult to interpret when the number of clusters is too high (Christianto et al., 2023). While Hierarchical Clustering, can provide a priority order of factors that affects satisfaction (Tosida et al., 2021), hierarchical clustering can be computationally more expensive or slower than methods like K-Means (Zare & Emadi, 2020). And while K-Medoid Clustering, can be better than K-Means since it is superior in terms of text processing and statistical grouping, it has shown to be less optimal than K-Means in terms of the quality of clustering results (Tosida et al., 2021).

For this report, K-Means will be used as it is more simple to implement and it is scalable with large datasets (Sakalessy & Purnomo, 2024). For the data that will be used in the report, data regarding demographics/job information, performance metrics, recognitions, and satisfaction/engagement indicators will be used (Christianto et al., 2023). The result will group data into clusters based on characteristics (performance, satisfaction levels, etc.. This can help Xero in identifying improvements for each cluster. For example, instead of implementing company-wide initiatives that may not resonate with all employees, the cluster analysis allows for the development of customized interventions and strategies for specific employee segments (Mensouri et al., 2022).

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Conclusion

In Xero, the use of business intelligence can impact the organization in many ways. First is using visualizations to provide interactive dashboards to display employee performance. This can be used to incentivize effort and performance, improve workforce management, productivity, employee engagement, and optimize work balance. Second is using prediction to predict employee attrition. This can be used provide early detection and

allow Xero to intervene proactively. Third is the use of clustering to improve employee engagement and satisfaction. This can be used to group employees into clusters and implement strategies for each specific segments to improve employee engagement and satisfaction.

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